

Abstract

MODEL CONTEXT PROTOCOL (MCP): ARCHITECTURE AND CONTEXTUAL INTEROPERABILITY FOR AI SYSTEMS

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Large Language Models (LLMs) are increasingly deployed as intelligent agents capable of interacting with external tools, databases, APIs, and software systems to perform complex real-world tasks. Without a unified communication standard, each integration historically required custom point-to-point implementations, resulting in fragmented and unmaintainable AI ecosystems. The Model Context Protocol (MCP), introduced by Anthropic in 2024, resolves this through an open, JSON-RPC-based standard that provides secure, interoperable, and bidirectional communication between AI models and external resources via a structured host–client–server architecture.

This seminar presents a comprehensive study of MCP, examining its core components—clients, servers, tools, resources, prompts, and sampling—and demonstrating how they enable AI systems to efficiently discover and interact with external services. It further explores MCP’s role in enterprise API interoperability and real-world IoT applications, including environmental monitoring systems. The study synthesises findings from ten peer-reviewed papers (2025–2026) spanning top venues including IEEE S&P, ACM, SPIE, Elsevier, IJCESEN, and IEEE Xplore, employing empirical analysis, large-scale static analysis of 1,899 open-source MCP servers, adversarial red-teaming, and applied system evaluations.

MCP demonstrably reduces integration complexity through its $N \times M$ connection model, cutting deployment time and lowering token overhead. However, large-scale studies expose critical security vulnerabilities—Tool Poisoning, Puppet attacks, Rug-Pull, and Parasitic Toolchain Attacks—spanning 12,230 tools and 1,360 servers. Analysis of open-source MCP servers further reveals prevalent code smells, bug patterns, and maintainability deficiencies that compound security risks. Practical auditing frameworks such as MCPSafetyScanner detect a significant portion of these threats, yet governance and compliance challenges persist. MCP represents a foundational protocol for scalable, interoperable AI agent ecosystems; realising its full potential demands standardised security controls, rigorous server auditing, and clear governance frameworks, with future research encompassing dynamic trust verification, formalised sandboxing, and cross-platform compliance standards.

References

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